ECO310 Midterm 1: Intuitive Study Guide

1 Consumer Preferences: What People Want

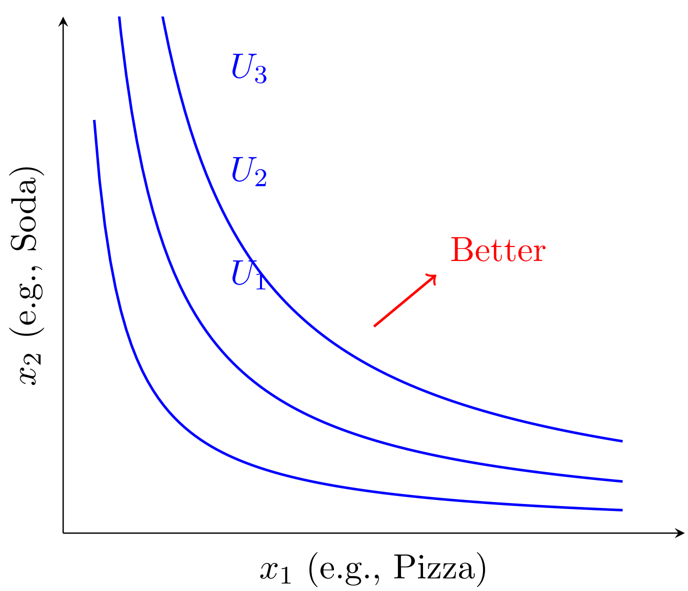
1.1 The Big Picture

Economics starts with a simple question: *How do people choose?* We need to model how consumers rank different bundles of goods.

Intuition

Think of it like ranking your favorite meals. You might prefer pizza over salad, salad over plain bread, etc. If your preferences are "rational," you can consistently rank any two options and those rankings don't contradict each other.

1.2 Indifference Curves: Your Preference Map

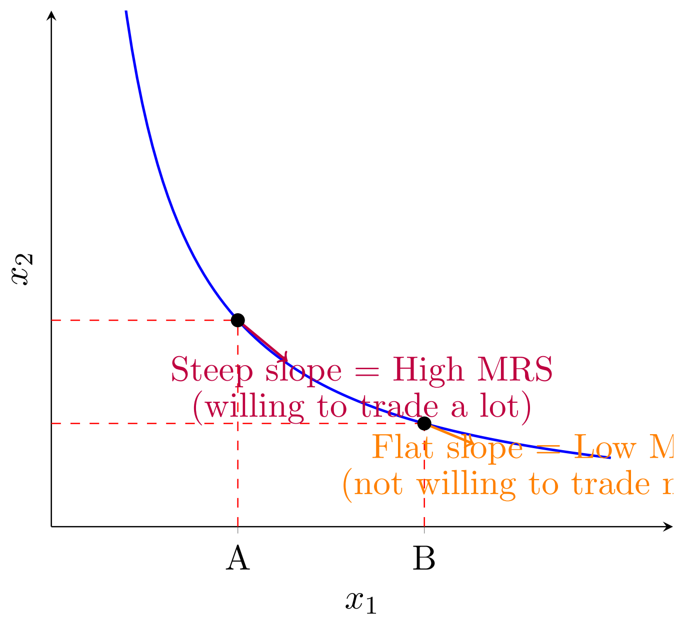


Intuition

An indifference curve shows all bundles that give you the **same happiness**. Higher curves = more happiness. The curves are smooth (no jumps) and bow inward (you like variety - having some of both goods is better than lots of one).

1.3 MRS: The Trade-Off Rate

Marginal Rate of Substitution (MRS) = How many units of x_2 you'd give up for one more x_1, while staying equally happy.



Formula: $MRS = \frac{MU_1}{MU_2}$ (ratio of marginal utilities)

Intuition

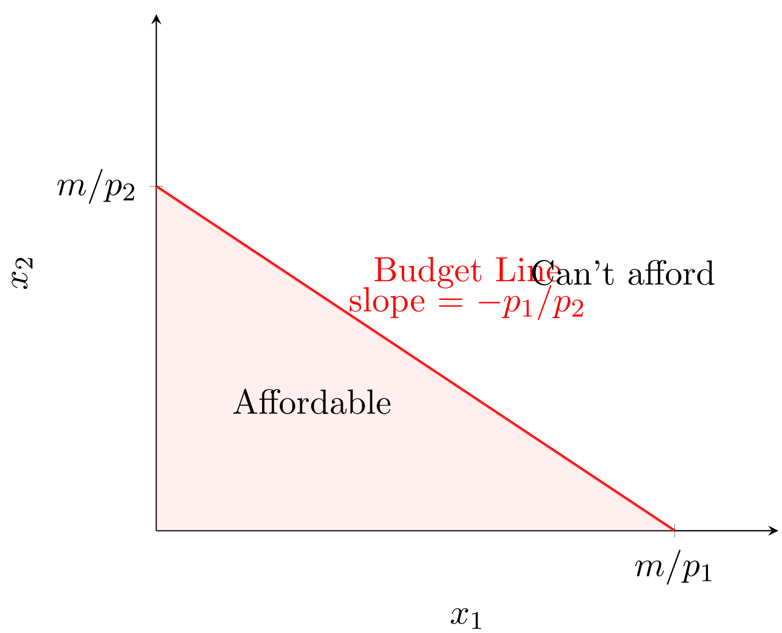
At point A, you have little pizza and lots of soda. You'd happily trade lots of soda for one more pizza (steep slope). At point B, you have more pizza, so you're less desperate - you'd only give up a little soda (flat slope). This is **diminishing MRS**.

2 Budget Constraints: What You Can Afford

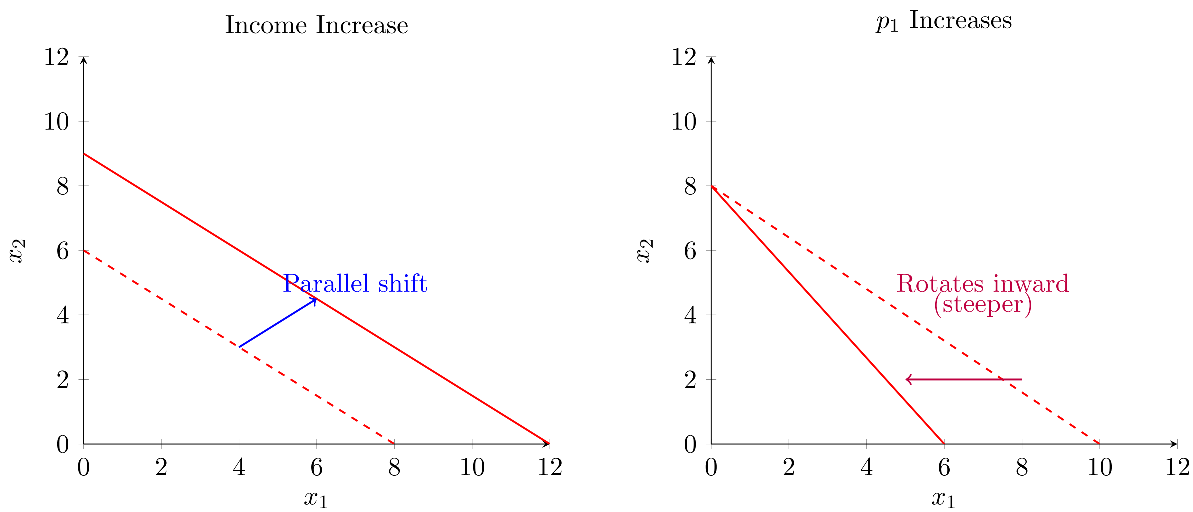
2.1 The Budget Line

You have income m and face prices p_1, p_2. Your budget constraint:

$$p_1x_1 + p_2x_2 = m$$



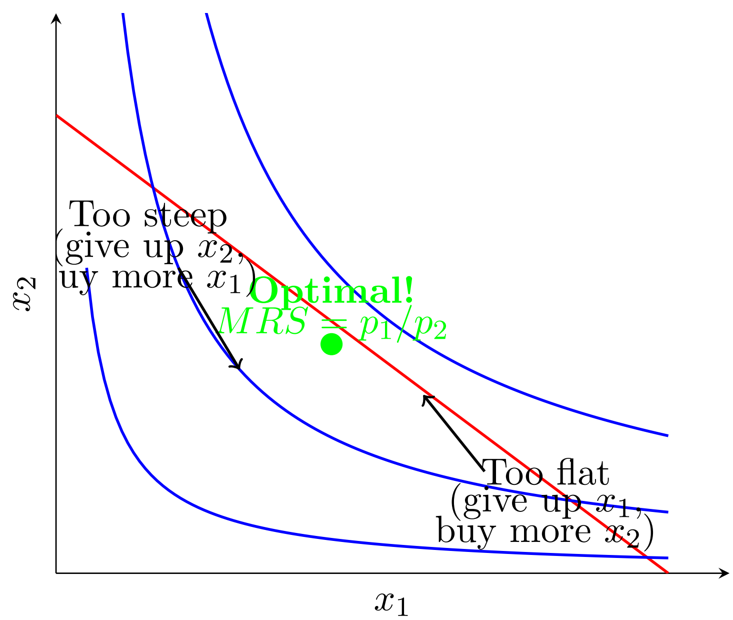
2.2 What Happens When Things Change?



3 Optimal Choice: Where to Shop

3.1 The Tangency Condition

Your goal: Reach the highest indifference curve you can afford.



How to Solve

How to find optimal consumption:

1. Write down utility function and budget constraint
2. Calculate MRS = MU_1/MU_2 by taking derivatives
3. Set $MRS = p_1/p_2$ (tangency condition)
4. Plug into budget constraint: $p_1x_1 + p_2x_2 = m$
5. Solve the two equations for x_1^* and x_2^*
6. Check: Are both positive? If not, you have a corner solution!

Example

Problem: $u = x_1^{0.5}x_2^{0.5}$, $p_1 = 2$, $p_2 = 4$, $m = 120$
Solution:

1. $MU_1 = 0.5x_1^{-0.5}x_2^{0.5}$, $MU_2 = 0.5x_1^{0.5}x_2^{-0.5}$
2. $MRS = \frac{MU_1}{MU_2} = \frac{x_2}{x_1}$
3. Set $\frac{x_2}{x_1} = \frac{p_1}{p_2} = \frac{2}{4} = \frac{1}{2}$
4. So $x_2 = \frac{1}{2}x_1$
5. Plug into budget: $2x_1 + 4(\frac{1}{2}x_1) = 120$
6. $4x_1 = 120 \Rightarrow x_1^* = 30$
7. $x_2^* = \frac{1}{2}(30) = 15$

Answer: Buy 30 units of good 1 and 15 units of good 2.

4 Special Utility Functions: Shortcuts

4.1 Cobb-Douglas: The "Spend a Fraction" Rule

$$u = x_1^a x_2^b$$

Intuition

This is the most common utility function. The magic rule: spend fraction $\frac{a}{a+b}$ of your income on good 1, and $\frac{b}{a+b}$ on good 2. **Always!** (Doesn't matter what prices are!)

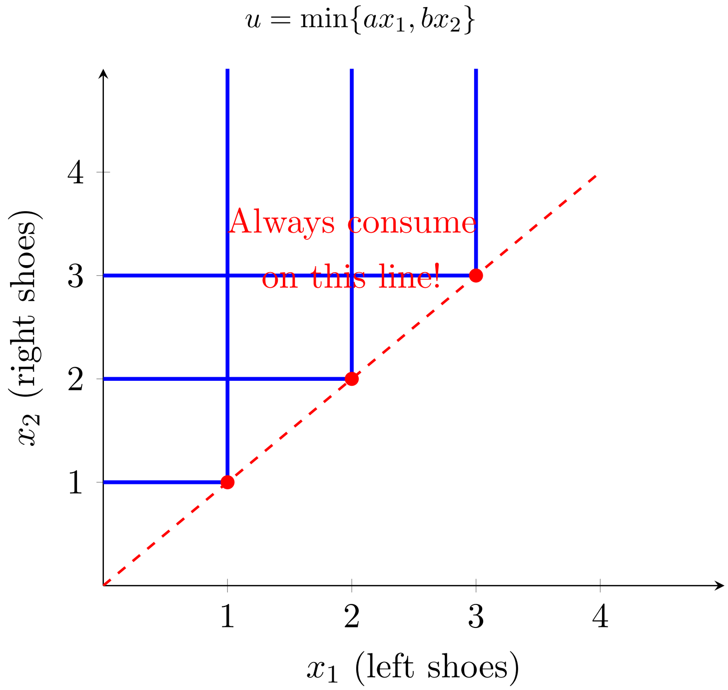
Demand formulas:

$$x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$$

Example

$u = x_1^{0.3} x_2^{0.7}$, any prices, income $m = 100$
Spend $\frac{0.3}{1.0} = 30\%$ on good 1, 70% on good 2.
If $p_1 = 5$: $x_1^* = \frac{30}{5} = 6$
If $p_2 = 2$: $x_2^* = \frac{70}{2} = 35$

4.2 Perfect Complements: The "Bundle" Rule



Intuition

Goods are consumed in fixed proportions (like left and right shoes). Extra of one is useless. The L-shaped curves show you don't care about having 5 left shoes if you only have 1 right shoe.

How to Solve

How to solve:

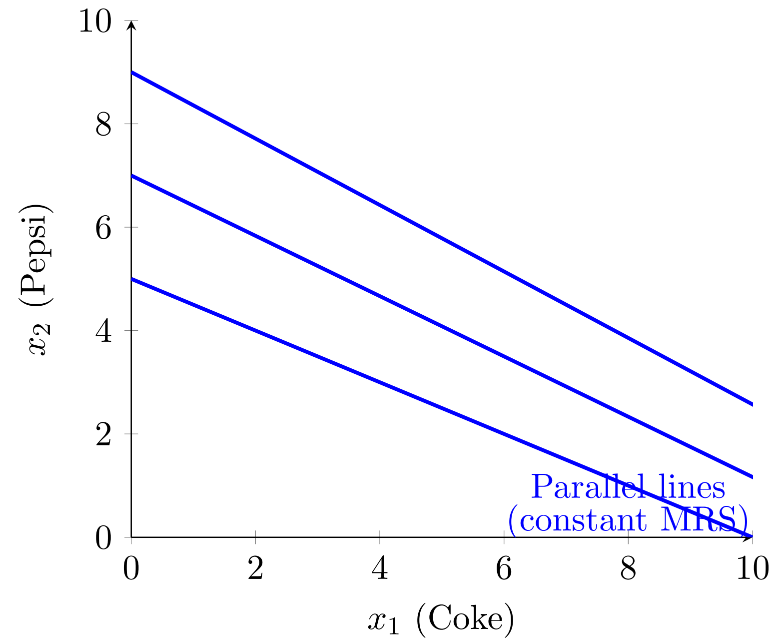
1. Always consume at the kink: $ax_1 = bx_2$
2. Plug into budget: $p_1 x_1 + p_2 x_2 = m$
3. Solve for x_1^* and x_2^*

Example

$u = \min\{x_1, 2x_2\}$ (need 2 units of x_2 for each x_1)
At kink: $x_1 = 2x_2$, so $x_2 = \frac{x_1}{2}$
Budget: $p_1 x_1 + p_2(\frac{x_1}{2}) = m$
$x_1^*(p_1 + \frac{p_2}{2}) = m$
$x_1^* = \frac{m}{p_1 + p_2/2}$, $x_2^* = \frac{m}{2p_1 + p_2}$

4.3 Perfect Substitutes: The "All or Nothing" Rule

$$u = ax_1 + bx_2$$



Intuition

Goods are perfect substitutes (like Coke vs Pepsi to someone who doesn't care). The indifference curves are straight lines. You'll buy ALL of whichever is cheaper per unit of utility.

How to Solve

Decision rule:

- If $\frac{a}{p_1} > \frac{b}{p_2}$: good 1 gives more "bang per buck" $\Rightarrow$ buy only $x_1 = m/p_1$
- If $\frac{a}{p_1} < \frac{b}{p_2}$: good 2 is better $\Rightarrow$ buy only $x_2 = m/p_2$
- If equal: you're indifferent, buy any combination on the budget line

4.4 Quasilinear: The "No Income Effect" Function

$$u = v(x_1) + x_2$$

Intuition

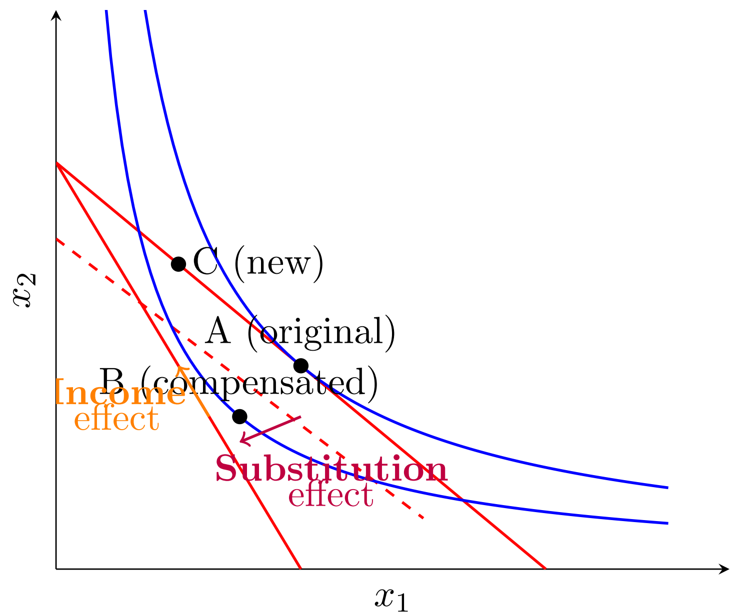
Good 2 is like "money" - you always want more of it, but it doesn't affect how you value good 1. Your demand for good 1 depends **only on its price**, not your income (as long as you have enough income to afford some of both goods).

The indifference curves are vertical translations of each other.

5 Income & Substitution Effects: Why Demand Changes

When price changes, two things happen:

5.1 The Decomposition



Intuition

When p_1 rises:

- **Substitution effect (A→B):** Good 1 is now relatively more expensive, so you substitute toward good 2. Move along the IC to the new price ratio. This ALWAYS reduces x_1.
- **Income effect (B→C):** You're effectively poorer (your money doesn't stretch as far). If x_1 is a normal good, you buy less. If it's inferior, you might buy more!

How to Solve

How to calculate:

1. Find original bundle at (p_1, p_2, m): call it A
2. Calculate utility at A: $\bar{u} = u(A)$
3. Find compensated bundle: minimize spending to reach $\bar{u}$ at new prices (p'_1, p_2, m). Call it B.
4. Substitution effect $= B - A$
5. Find new actual bundle at (p'_1, p_2, m): call it C
6. Income effect $= C - B$
7. Total effect $= C - A =$ sub + income

Common Mistake

For **perfect complements**, there is NO substitution effect (you can't substitute - you need them in fixed proportions). The entire change is income effect!
For **quasilinear**, there is NO income effect for the quasilinear good (at interior solutions).

6 Uncertainty & Risk

6.1 Expected Utility Theory

You face a lottery with outcomes $c_1, c_2, \ldots$ with probabilities $p_1, p_2, \ldots$

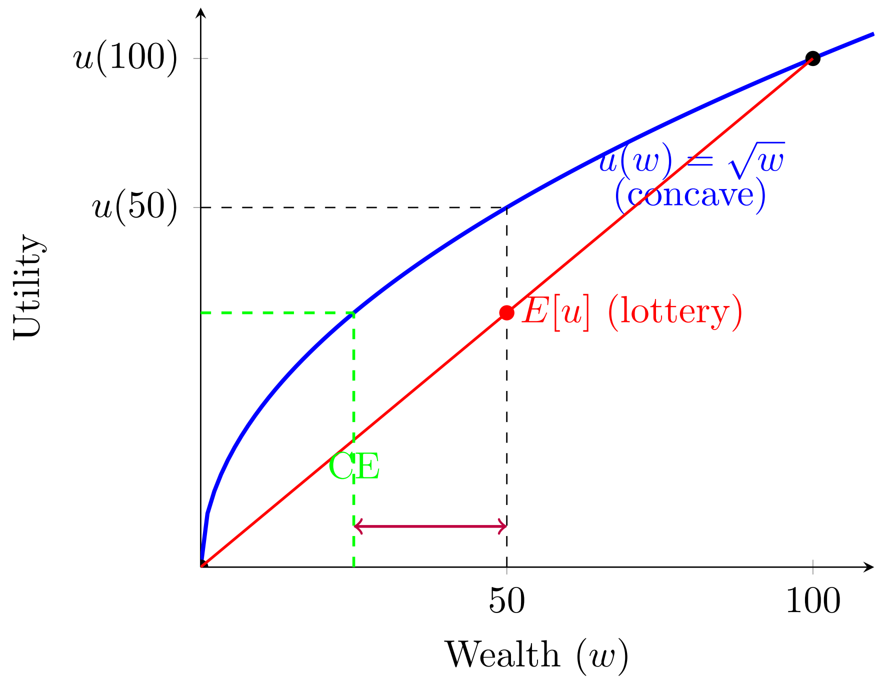
Expected Utility:

$$EU = p_1 u(c_1) + p_2 u(c_2) + \cdots$$

Intuition

You don't just care about the average outcome - you care about the utility of each outcome. If you're risk-averse, bad outcomes hurt more than good outcomes help (in utility terms).

6.2 Risk Aversion



Intuition

Risk averse: Your utility function is concave (bends down). This means:

- You prefer \$50 for sure over a 50-50 gamble of \$0 or \$100
- Even though the expected value is the same (\$50), the expected utility of the gamble is lower
- You'd pay to avoid risk (insurance!)

Key formulas:

- **Certainty Equivalent (CE):** The sure amount worth the same as the gamble: $u(CE) = E[u]$
- **Risk Premium (RP):** How much you'd pay to avoid the risk: $RP = E[w] - CE$

Example

Problem: You have $u(w) = \sqrt{w}$. You face a 50-50 lottery of \$0 or \$100.
Solution:

1. Expected value: $E[w] = 0.5(0) + 0.5(100) = 50$
2. Expected utility: $E[u] = 0.5\sqrt{0} + 0.5\sqrt{100} = 0 + 5 = 5$
3. Certainty equivalent: $u(CE) = 5 \Rightarrow \sqrt{CE} = 5 \Rightarrow CE = 25$
4. Risk premium: $RP = 50 - 25 = 25$

Interpretation: The lottery is worth only \$25 to you (even though it has an expected value of \$50). You'd pay up to \$25 to avoid this risk!

6.3 Insurance Problem

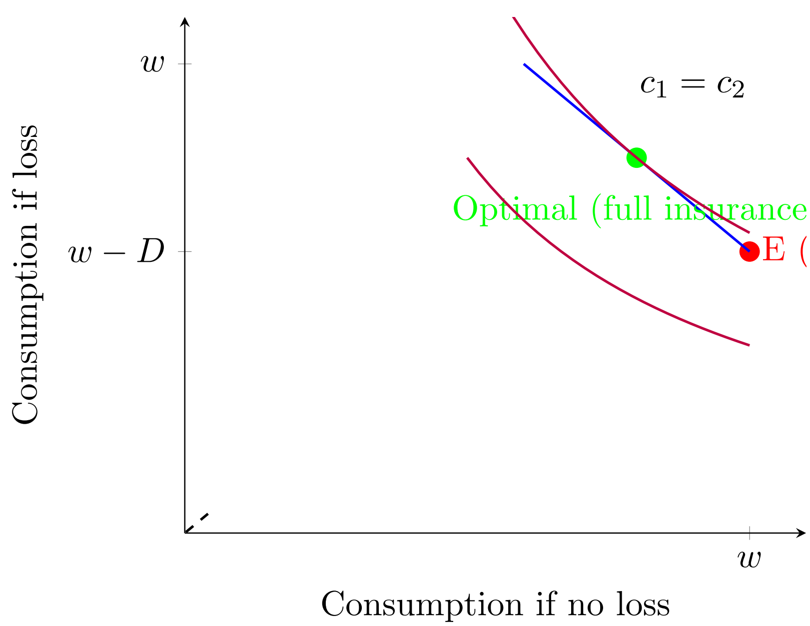
You have wealth w and might lose D with probability π. Insurance costs q per dollar of coverage K.

Your wealth in each state:

- No loss (prob $1 - \pi$): $w - qK$
- Loss (prob π): $w - D - qK + K = w - D + K(1 - q)$

Expected utility:

$$EU = (1 - \pi)u(w - qK) + \pi u(w - D + K(1 - q))$$



Intuition

With fair insurance ($q = \pi$): A risk-averse person buys full insurance ($K = D$). This makes consumption the same in both states - no uncertainty! The budget line is drawn so it passes through the 45° line, where you're fully insured.
With unfair insurance ($q > \pi$): You buy partial insurance - some, but not full, coverage.

How to Solve

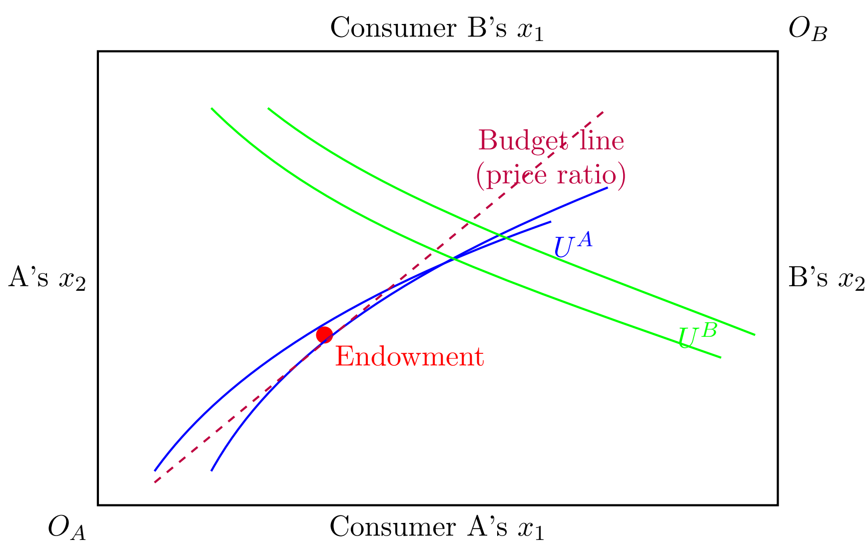
How to solve:

1. Write expected utility as function of K
2. Take derivative: $\frac{dEU}{dK} = 0$
3. This gives: $(1-\pi)u'(w-qK)\cdot(-q) + \pi u'(w-D+K(1-q))\cdot(1-q) = 0$
4. Simplify to find K^*
5. For fair insurance, this yields $K^* = D$ (full insurance)

7 General Equilibrium: Markets Clear

7.1 Edgeworth Box

Two consumers (A and B), two goods (1 and 2), total endowments (E_1, E_2).



Intuition

The Edgeworth box shows all possible allocations. The width is total x_1, the height is total x_2. A reads from the bottom-left corner, B reads from the top-right (flipped). Any point in the box is a feasible allocation (adding up to total endowments).

7.2 Competitive Equilibrium

Equilibrium conditions:

1. Each consumer maximizes utility given budget: $p \cdot x^i = p \cdot e^i$
2. Markets clear: $x_1^A + x_1^B = E_1$ and $x_2^A + x_2^B = E_2$

How to Solve

How to solve:

1. Normalize $p_2 = 1$ (so all prices are relative to good 2)
2. Calculate each person's income: $m^A = p_1 e_1^A + e_2^A$ (value of their endowment)
3. Solve each person's UMP to get demands: $x_1^A(p_1)$ and $x_1^B(p_1)$
4. Market clearing: $x_1^A(p_1) + x_1^B(p_1) = E_1$
5. Solve for equilibrium price p_1^*
6. Plug back to get equilibrium allocations
7. Verify: $x_2^A, x_2^B \geq 0$ (no one demands negative amounts)

Example

Problem: 2 consumers with Cobb-Douglas utilities:

- A: $u^A = x_1^{0.5} x_2^{0.5}$, endowment $(4, 0)$
- B: $u^B = x_1^{0.5} x_2^{0.5}$, endowment $(0, 4)$

Solution:

1. Set $p_2 = 1$, so prices are $(p_1, 1)$
2. Incomes: $m^A = 4p_1$, $m^B = 4$
3. CD demands: spend half on each good
   - $x_1^A = \frac{m^A/2}{p_1} = \frac{2p_1}{p_1} = 2$
   - $x_1^B = \frac{m^B/2}{p_1} = \frac{2}{p_1}$
4. Market clearing: $2 + \frac{2}{p_1} = 4$
5. $\frac{2}{p_1} = 2 \Rightarrow p_1^* = 1$
6. Equilibrium: $(p_1, p_2) = (1, 1)$
7. Allocations: $x^A = (2, 2)$, $x^B = (2, 2)$

They split everything equally! (Makes sense - identical preferences, symmetric endowments.)

7.3 Pareto Efficiency

An allocation is **Pareto efficient (PE)** if you can't make someone better off without making someone else worse off.

Intuition

At a PE allocation, the indifference curves of A and B are tangent: $MRS^A = MRS^B$. If they weren't tangent, there would be a "lens" between the curves where both could be better off.

Key Results:

- **First Welfare Theorem:** Every competitive equilibrium is Pareto efficient
- **Second Welfare Theorem:** Any Pareto efficient allocation can be achieved as a competitive equilibrium (with the right redistribution of endowments)

8 Quick Problem-Solving Guide

How to Solve

For utility maximization:

1. Identify the utility function type (CD, PC, PS, QL, or general)
2. If CD: use the shortcut (spend fraction $a/(a+b)$ on each good)
3. If PC: set $ax_1 = bx_2$ and use budget
4. If PS: compare a/p_1 vs b/p_2 and buy all of the better one
5. Otherwise: $MRS = p_1/p_2$ and budget constraint
6. Always check: are your answers non-negative? If not, corner solution!

How to Solve

For income/substitution effects:

1. Original bundle (A) at old prices
2. Calculate original utility
3. Compensated bundle (B): new prices, old utility (minimize cost or stay on same IC)
4. New bundle (C) at new prices
5. Sub effect $= B - A$ (always negative for own price)
6. Income effect $= C - B$ (sign depends on normal/inferior)

How to Solve

For expected utility:

1. List all outcomes and probabilities
2. Write $EU = \sum p_i u(c_i)$
3. If finding optimal choice (insurance, portfolio): take FOC, set = 0, solve
4. If finding CE: set $u(CE) = EU$ and solve for CE
5. If finding RP: $RP = E[w] - CE$

Common Mistake

Common traps:

- Forgetting to check $x_i \geq 0$ (corner solutions!)
- Mixing up substitution effect direction (always buy LESS of the good that got more expensive)
- Using arbitrary transforms on vNM utility (only affine transforms allowed!)
- Confusing variance (σ^2) with std deviation (σ)
- Forgetting that PC has zero substitution effect